Lab 7: Fuzzing for Input Validation Bugs

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Forked repository: https://github.com/varshinigt/L6_L7_28_airline_baggage/tree/main/lab7

Step 1: Installing dependencies (pip install hypothesis pytest, inside a virtual environment)

```
(lab7-venv) Varshini@Varshinis-MacBook-Air lab7 % pip install hypothesis pytest
Requirement already satisfied: hypothesis in /Users/Varshini/lab7-venv/lib/python3.15/site-packages (6.170.15)
Requirement already satisfied: pytest in /Users/Varshini/lab7-venv/lib/python3.15/site-packages (9.2.0)
Requirement already satisfied: sortedcontainers<3.0.0,>=2.1.0 in /Users/Varshini/lab7-venv/lib/python3.15/site-packages (from hypothesis) (2.4.0)
Requirement already satisfied: iniconfig>=2 in /Users/Varshini/lab7-venv/lib/python3.15/site-packages (from pytest) (2.3.1)
Requirement already satisfied: packaging>=24 in /Users/Varshini/lab7-venv/lib/python3.15/site-packages (from pytest) (26.3)
Requirement already satisfied: pluggy<2,>=1.6 in /Users/Varshini/lab7-venv/lib/python3.15/site-packages (from pytest) (1.7.0)
Requirement already satisfied: pygments>=2.15 in /Users/Varshini/lab7-venv/lib/python3.15/site-packages (from pytest) (2.21.0)

[notice] A new release of pip is available: 26.2.2 -> 26.3
[notice] To update, run: pip install --upgrade pip
(lab7-venv) Varshini@Varshinis-MacBook-Air lab7 % pip show hypothesis pytest | grep -E "Name|Version"
Name: hypothesis
Version: 6.170.15
Name: pytest
Version: 9.2.0
(lab7-venv) Varshini@Varshinis-MacBook-Air lab7 %
```

Step 2: Running the buggy application (crashes with ValueError on input "hello")

```
(lab7-venv) Varshini@Varshinis-MacBook-Air lab7 % python3 airline_baggage.py
==== Running parse_booking ====
Enter a booking code (e.g. AI-203-12A): hello
Traceback (most recent call last):
  File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/airline_baggage.py", line 29, in <module>
    main()
    ~~~~^^
  File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/airline_baggage.py", line 15, in main
    result = parse_booking(user_input)
  File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/airline_baggage.py", line 2, in parse_booking
    airline, flight, seat = code.split('-')
    ^^^^^^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 3, got 1)
(lab7-venv) Varshini@Varshinis-MacBook-Air lab7 %
```

Step 3: Completing test_airline_baggage.py with Hypothesis (@given(st.text() | st.none()) for all three functions)

```
(lab7-venv) Varshini@Varshinis-MacBook-Air lab7 % cat > test_airline_baggage.py << 'EOF'
from hypothesis import given, strategies as st
from airline_baggage import parse_booking, total_weight, tag_label

@given(st.text() | st.none())
def test_parse_booking_no_crash(s):
    try:
        parse_booking(s)
    except Exception as e:
        assert False, f"parse_booking crashed with: {e!r}"

@given(st.text() | st.none())
def test_total_weight_no_crash(s):
    try:
        total_weight(s)
    except Exception as e:
        assert False, f"total_weight crashed with: {e!r}"

@given(st.text() | st.none())
def test_tag_label_no_crash(s):
    try:
        tag_label(s)
    except Exception as e:
        assert False, f"tag_label crashed with: {e!r}"
EOF
```

Step 4: Running Hypothesis fuzz tests on the buggy code (3 failed) - part 1

```
(lab7-venv) Varshini@Varshinis-MacBook-Air lab7 % cat > test_airline_baggage.py << 'EOF'
from hypothesis import given, strategies as st
from airline_baggage import parse_booking, total_weight, tag_label

@given(st.text() | st.none())
def test_parse_booking_no_crash(s):
    try:
        parse_booking(s)
    except Exception as e:
        assert False, f"parse_booking crashed with: {e!r}"

@given(st.text() | st.none())
def test_total_weight_no_crash(s):
    try:
        total_weight(s)
    except Exception as e:
        assert False, f"total_weight crashed with: {e!r}"

@given(st.text() | st.none())
def test_tag_label_no_crash(s):
    try:
        tag_label(s)
    except Exception as e:
        assert False, f"tag_label crashed with: {e!r}"
EOF
(lab7-venv) Varshini@Varshinis-MacBook-Air lab7 % clear

(lab7-venv) Varshini@Varshinis-MacBook-Air lab7 % python3 -m pytest test_airline_baggage.py
============================= test session starts ==============================
platform darwin -- Python 3.15.2, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7
plugins: hypothesis-6.170.15
collected 3 items

test_airline_baggage.py FFF                                              [100%]

=================================== FAILURES ===================================
_____________________________ test_parse_booking_no_crash ______________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 5, in test_parse_booking_no_crash
  |     def test_parse_booking_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 7, in test_parse_booking_no_crash
    |     parse_booking(s)
    |     ~~~~~~~~~~~~~^^^
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/airline_baggage.py", line 2, in parse_booking
    |     airline, flight, seat = code.split('-')
    |     ^^^^^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 3, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 9, in test_parse_booking_no_crash
    |     assert False, f"parse_booking crashed with: {e!r}"
    | AssertionError: parse_booking crashed with: ValueError('not enough values to unpack (expected 3, got 1)')
    | assert False
    | Falling test case: test_parse_booking_no_crash(
    |     s='',  # or any other generated value
    | )
```

Step 4 (continued) - part 2

```
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 7, in test_parse_booking_no_crash
    |     parse_booking(s)
    |     ~~~~~~~~~~~~~^^^
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/airline_baggage.py", line 2, in parse_booking
    |     airline, flight, seat = code.split('-')
    |                             ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 9, in test_parse_booking_no_crash
    |     assert False, f"parse_booking crashed with: {e!r}"
    | AssertionError: parse_booking crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_parse_booking_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________________________________ test_total_weight_no_crash _____________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 12, in test_total_weight_no_crash
  |     def test_total_weight_no_crash(s):
  |                                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 14, in test_total_weight_no_crash
    |     total_weight(s)
    |     ~~~~~~~~~~~~^^^
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/airline_baggage.py", line 6, in total_weight
    |     return sum(float(w) for w in weights_csv.split(','))
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/airline_baggage.py", line 6, in <genexpr>
    |     return sum(float(w) for w in weights_csv.split(','))
    |                ~~~~~^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 16, in test_total_weight_no_crash
    |     assert False, f"total_weight crashed with: {e!r}"
    | AssertionError: total_weight crashed with: ValueError("could not convert string to float: ''")
    | assert False
    | Failing test case: test_total_weight_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |     /Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py:15
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 14, in test_total_weight_no_crash
    |     total_weight(s)
    |     ~~~~~~~~~~~~^^^
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/airline_baggage.py", line 6, in total_weight
    |     return sum(float(w) for w in weights_csv.split(','))
    |                              ^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
```

Step 4 (continued) - part 3

```
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 16, in test_total_weight_no_crash
    |     assert False, f"total_weight crashed with: {e!r}"
    | AssertionError: total_weight crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_total_weight_no_crash(
    |     s=None,  # or any other generated value
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py:15
    +------------------------------------
_____________________________________________________ test_tag_label_no_crash _____________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 19, in test_tag_label_no_crash
  |     def test_tag_label_no_crash(s):
  |                                 ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 21, in test_tag_label_no_crash
    |     tag_label(s)
    |     ~~~~~~~~~^^^
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/airline_baggage.py", line 10, in tag_label
    |     return parts[0][0].upper() + '. ' + parts[1].upper()
    |            ~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 23, in test_tag_label_no_crash
    |     assert False, f"tag_label crashed with: {e!r}"
    | AssertionError: tag_label crashed with: IndexError('list index out of range')
    | assert False
    | Failing test case: test_tag_label_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py:22
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 21, in test_tag_label_no_crash
    |     tag_label(s)
    |     ~~~~~~~~~^^^
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/airline_baggage.py", line 9, in tag_label
    |     parts = name.split()
    |             ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 23, in test_tag_label_no_crash
    |     assert False, f"tag_label crashed with: {e!r}"
    | AssertionError: tag_label crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_tag_label_no_crash(
    |     s=None,
    | )
```

Step 4 (continued) - part 4: summary, 3 failed

```
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py:15
    +------------------------------------
_____________________________________________________ test_tag_label_no_crash _____________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 19, in test_tag_label_no_crash
  |     def test_tag_label_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 21, in test_tag_label_no_crash
    |     tag_label(s)
    |     ~~~~~~~~~^^^
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/airline_baggage.py", line 10, in tag_label
    |     return parts[0][0].upper() + '. ' + parts[1].upper()
    |                ~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 23, in test_tag_label_no_crash
    |     assert False, f"tag_label crashed with: {e!r}"
    | AssertionError: tag_label crashed with: IndexError('list index out of range')
    | assert False
    | Failing test case: test_tag_label_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py:22
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 21, in test_tag_label_no_crash
    |     tag_label(s)
    |     ~~~~~~~~~^^^
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/airline_baggage.py", line 9, in tag_label
    |     parts = name.split()
    |             ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py", line 23, in test_tag_label_no_crash
    |     assert False, f"tag_label crashed with: {e!r}"
    | AssertionError: tag_label crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_tag_label_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/Varshini/Desktop/L6_L7_28_airline_baggage/lab7/test_airline_baggage.py:22
    +------------------------------------
=========================================================== short test summary info ============================================================
FAILED test_airline_baggage.py::test_parse_booking_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_airline_baggage.py::test_total_weight_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_airline_baggage.py::test_tag_label_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================================================ 3 failed in 3.02s =============================================================
(lab7-venv) Varshini@Varshinis-MacBook-Air lab7 %
```

Step 5: Running the fixed application (invalid inputs handled, valid inputs still work)

```
(lab7-venv) Varshini@Varshinis-MacBook-Air lab7 % python3 airline_baggage.py
==== Running parse_booking ====
Enter a booking code (e.g. AI-203-12A): hello
Invalid booking code.

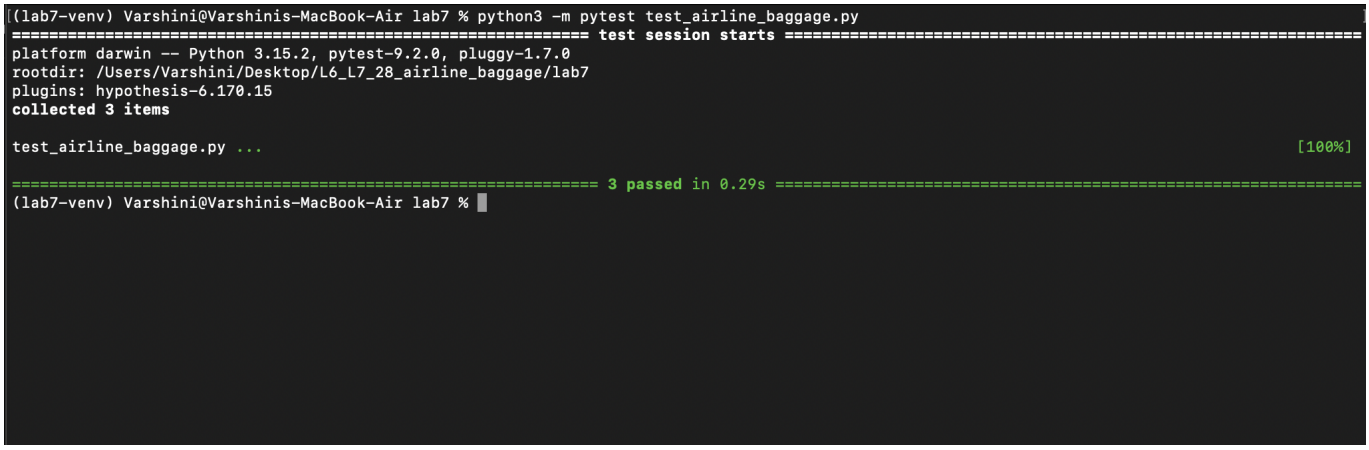
==== Running total_weight ====
Enter bag weights in kg as CSV (e.g. 12.5,7,23): abc,5
Invalid weights. Use non-negative numbers separated by commas.

==== Running tag_label ====
Enter a passenger name (first and last): John
Invalid name. Enter both first and last name.
(lab7-venv) Varshini@Varshinis-MacBook-Air lab7 % python3 airline_baggage.py
==== Running parse_booking ====
Enter a booking code (e.g. AI-203-12A): AI-203-12A
Airline, Flight and Seat Letter: ('AI', 203, 'A')

==== Running total_weight ====
Enter bag weights in kg as CSV (e.g. 12.5,7,23): 12.5,7,23
Total Weight: 42.5

==== Running tag_label ====
Enter a passenger name (first and last): John Smith
Tag Label: J. SMITH
(lab7-venv) Varshini@Varshinis-MacBook-Air lab7 %
```

Step 6: Re-running the fuzz tests on the fixed code (3 passed)



Bugs found: empty string and None caused ValueError / IndexError / AttributeError in all three functions (wrong number of "-" parts, empty weight entries, names with fewer than two words, non-string input).

Fixes: type checks, input validation and try/except; invalid input returns None and main() prints a friendly message.